

Lab Handouts (CSE 103)

Topic: Loop

Lab Objective:

- Familiarize students with loops in the C programming language.

Lab Outcome:

After completing this lab successfully, students will be able to:

- Understand** the concepts of loops.
- Demonstrate** the skill to write C programs to solve problems with modest complexity using loops.

Required Tools:

Code::Blocks is a free, open-source, cross-platform IDE that supports multiple compilers including GCC, Clang and Visual C++. It is developed in C++ using wxWidgets as the GUI toolkit. Using a plugin architecture, its capabilities and features are defined by the provided plugins. Currently, Code::Blocks is oriented towards C, C++, and Fortran.

Problem 1: Write a loop to print up to n integer (positive) numbers starting from 1.

Solution:

Using while loop	Using for loop	Using do-while loop
<pre>int n, i; scanf("%d", &n); i=1; while(i<=n){ printf("%d\n", i); i++; }</pre>	<pre>scanf("%d", &n); for(i=1; i<=n; i++){ printf("%d\n", i); }</pre>	<pre>int n, i; scanf("%d", &n); i=1; do{ printf("%d\n", i); i++; }while(i<=n);</pre>

Description: Try to run the program and put an integer number as input. Try to understand the output and how the workflow works.

A few sample input-output:

Sample Input	Sample Output
3	1 2 3

Problem 2: Write a C program to print all the even numbers from 1 to n in **descending** order.

Solution:

Solution 1	Solution 2
<pre>scanf("%d", &n); if(n%2==1) n--; while(n>0){ printf("%d\n", n); n-=2; }</pre>	<pre>scanf("%d", &n); if(n%2==1) n--; for(; n>0; n-=2){ printf("%d\n", n); }</pre>

A few sample input-output:

Sample Input	Sample Output
3	2
8	8 6 4 2

Problem 3: Write a C program that will take two integer numbers as input (x, y), where $1 \leq x, y \leq 26$ and $x \leq y$, and print the x^{th} letter to y^{th} letter in ascending order in English alphabet (uppercase only).

Solution:

Solution 1	Solution 2
<pre>scanf("%d %d", &x, &y); for(i=x ; i<=y ; i++){ printf("%c\n", 'A'+i-1); }</pre>	<pre>scanf("%d %d", &x, &y); i=x; while(i<=y){ printf("%c\n", 'A'+i-1); i++; }</pre>

A few sample input-output:

Sample Input	Sample Output
1 3	A B C
3 7	C D E F G

Problem 4: Write a C program to check if a number is Palindrome or not.

Solution:

Solution 1

```
scanf("%d", &n);
num= n;
reverse_num =0;
while(n>0){
    rem = n%10;
    reverse_num = reverse_num *10 +rem;
    n/=10; // n = n/10;
}
if(num==reverse_num){
    printf("Palindrome.\n");
}
else{
    printf("Not a Palindrome!");
}
```

A few sample input-output:

Sample Input	Sample Output
1221	Palindrome.
126221	Not a Palindrome!
123454321	Palindrome.

Problem 5: Write a C program that takes an integer number (n) as input and then it takes n pairs of integers numbers [(a₁, b₁), (a₂,b₂), (a_n,b_n)] as input and displays the sum of each pair of numbers.

Solution:

Solution 1

```
scanf("%d", &n);
for(i=0; i<n; i++){
    scanf("%d %d", &a, &b);
    printf("%d\n", a+b);
}
```

A few sample input-output:

Sample Input 1	Sample Output 1
3 1 5 5 5 7 8	6 10 15

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Problem 6: Write a C program that will take a pair of integer numbers (a and b) as input and will display the sum of those numbers. Your program should run until **End Of File** (EoF).

Solution:

Solution 1
<pre>while (scanf("%d %d", &a, &b) == 2) { printf("%d\n", a+b); }</pre>
Solution 2
<pre>while (scanf("%d %d", &a, &b) != EOF) { printf("%d\n", a+b); }</pre>

A few sample input-output:

Sample Input	Sample Output
2 3	5
5 10	15
10 10	20

Note: Press Ctrl+Z (for windows) or Ctrl+D (for linux) to terminate your program.

[**N.B.** This handout does not contain the complete code, rather it contains the code segment representing the main idea of the problem.]

Additional Problem List:

1. Write a program to input a number **N** and print its multiplication table up to **10** .
2. Write a program to input a number **N** and calculate its factorial .
3. Write a program to input a number and count how many digits it has .
4. Write a program to input a number and check if it's a prime number .
5. Write a program to input a number and calculate the sum of its digits
6. Write a program to input two numbers, base and exponent, and calculate the result of $\text{base}^{\text{exponent}}$.
7. Link: <https://www.w3resource.com/c-programming-exercises/while-loop/index.php>
8. Link: <https://codeforwin.org/c-programming/for-do-while-loop-programming-exercises> (Problem 1-26)